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|---------------------------------------------------------------------------|-------------------------------------------|
| Program: <b>Diploma in Artificial Intelligence &amp; Machine Learning</b> |                                           |
| Course Code: <b>3341</b>                                                  | Course Title: <b>Discrete Mathematics</b> |
| Semester: 3                                                               | Credits: <b>4</b>                         |
| Course Category: <b>Program Core</b>                                      |                                           |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>                                  | Periods per semester: <b>60</b>           |

### Course Objectives:

- Students will be able to develop logical thinking for solving computational problems.

### Course Prerequisites:

| Topic                                             | Course Title   | Course code | Semester |
|---------------------------------------------------|----------------|-------------|----------|
| Basic knowledge of algebra, geometry and calculus | Mathematics I  | 1001        | 1        |
|                                                   | Mathematics II | 2002        | 2        |

### Course Outcomes

On completion of the course, the students will be able to:

| CO <sub>n</sub> | Description                                                        | Duration (Hours) | Cognitive Level |
|-----------------|--------------------------------------------------------------------|------------------|-----------------|
| CO1             | Apply the basic concepts of mathematical logic to solve problems   | 16               | Applying        |
| CO2             | Demonstrate sets, relations and functions                          | 16               | Applying        |
| CO3             | Solve counting problems by applying elementary counting techniques | 16               | Applying        |
| CO4             | Summarize the abstract algebraic systems                           | 10               | Understanding   |
|                 | Series Test                                                        | 2                |                 |

## CO-PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             | 3   |     |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

## Course Outline

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                 | Description                                                             | Duration (Hours) | Cognitive Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Apply the basic concepts of mathematical logic to solve problems</b> |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                           | Describe the concept of propositional Logic                             | 2                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                           | Solve problems using truth tables and logical operations                | 3                | Applying        |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                           | Prove logical equivalence, contingency, tautology and contradictions    | 4                | Applying        |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                           | Describe the concept of predicate Logic                                 | 3                | Understanding   |
| M1.05                                                                                                                                                                                                                                                                                                                                                                                                           | Apply inference rules to solve problems                                 | 4                | Applying        |
| <b>Contents:</b><br>Mathematical logic:<br>Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction,<br>Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. |                                                                         |                  |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Demonstrate sets, relations and functions</b>                        |                  |                 |
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                           | Outline the basic principles of set                                     | 2                | Understanding   |

|                                                                                                                                                                                                                                                                                                                                                                               |                                                                                    |   |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---|---------------|
| M2.02                                                                                                                                                                                                                                                                                                                                                                         | Demonstrate the operations on sets                                                 | 2 | Applying      |
| M2.03                                                                                                                                                                                                                                                                                                                                                                         | Apply inclusion exclusion principle                                                | 2 | Applying      |
| M2.04                                                                                                                                                                                                                                                                                                                                                                         | Demonstrate the concept of relations                                               | 2 | Understanding |
| M2.05                                                                                                                                                                                                                                                                                                                                                                         | Identify the equivalence relations                                                 | 2 | Applying      |
| M2.06                                                                                                                                                                                                                                                                                                                                                                         | Illustrate the equivalence classes from equivalence relation                       | 3 | Applying      |
| M2.07                                                                                                                                                                                                                                                                                                                                                                         | Demonstrate the concept of functions                                               | 3 | Applying      |
|                                                                                                                                                                                                                                                                                                                                                                               | Series Test – I                                                                    | 1 |               |
| <b>Contents:</b><br><b>Set theory:-</b> Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion-exclusion principle<br><b>Relations:</b> Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class<br><b>Functions:</b> Mappings, Injection and Surjections, Composition of functions, Inverse functions |                                                                                    |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>Solve counting problems by applying elementary counting techniques</b>          |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                         | Solve counting problems by applying the techniques using the product and sum rules | 3 | Applying      |
| M3.02                                                                                                                                                                                                                                                                                                                                                                         | Solve counting problems by applying permutations & combinations                    | 3 | Applying      |
| M3.03                                                                                                                                                                                                                                                                                                                                                                         | Solve counting problems by applying the pigeon-hole principle                      | 3 | Applying      |
| M3.04                                                                                                                                                                                                                                                                                                                                                                         | Explain Binomial theorem                                                           | 2 | Understanding |
| M3.05                                                                                                                                                                                                                                                                                                                                                                         | Make use of recurrence relations to solve problems                                 | 3 | Applying      |
| M3.06                                                                                                                                                                                                                                                                                                                                                                         | Illustrate generating functions                                                    | 2 | Applying      |
| <b>Contents:</b><br>Basic counting principle, Rule of Sum, Rule of Product<br>Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof).<br>Recurrence relation, Generating functions                                                                                                                                                              |                                                                                    |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>Summarize the abstract algebraic systems</b>                                    |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                         | Differentiate homomorphism and isomorphism                                         | 2 | Understanding |

|                                                                                                                                                                                                               |                                                                                                   |   |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---|---------------|
| M4.02                                                                                                                                                                                                         | Describe the abstract algebraic systems - Groups, subgroups, Semigroups, Monoids & abelian groups | 5 | Understanding |
| M4.03                                                                                                                                                                                                         | Outline the concept of lattice by definition, properties                                          | 3 | Understanding |
|                                                                                                                                                                                                               | Series test II                                                                                    | 1 |               |
| <b>Contents:</b><br>Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems)<br><br>Lattice -Definition, properties, Types of lattice |                                                                                                   |   |               |

**Text / Reference:**

| T/R | Book Title / Author                                                                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------|
| T1  | Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson    |
| T2  | Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 |
| T3  | “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005                                             |

**Online Resources:**

| Sl. No. | Website Link                                                                                                                |
|---------|-----------------------------------------------------------------------------------------------------------------------------|
| 1       | <a href="https://archive.nptel.ac.in/courses/111/106/111106086/">https://archive.nptel.ac.in/courses/111/106/111106086/</a> |